SECTION 33 11 12
STEEL LINE PIPE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Line Pipe for BBN9 Turnout:

1. Payment: Lump sum price offered in Price Schedule 4.

a. Includes pipes, valves, and associated appurtenances shown on Drawings 1695-D-62548.

b. Includes cost of the following:

1) Pipe trench excavation in accordance with Section 31 23 22 – Pipe Trench Earthwork.

2) Furnishing and placing materials, including but not limited to the following:

a) Fittings, bedding, embedment, utility markers, concrete, reinforcing steel, blocking and collars, not included elsewhere.

b) Laying pipe.

B. Line Pipe for Coyote Canyon

1. Payment: Lump sum price offered in Price Schedule 5.

a. Includes cost of the following:

1) Pipe trench excavation in accordance with Section 31 23 22 – Pipe Trench Earthwork.

2) Furnishing and installing Coyote Canyon Turnout as shown on Drawing 1695-D-62550, including but not limited to the following:

a) Fittings, bedding, embedment, utility markers, concrete, reinforcing steel, blocking and collars, not included elsewhere.

b) Laying pipe.

C. Does not include pipe specified in Section 35 21 95 – Steel Manifold Piping.

1.02 PROJECT CONDITIONS

A. Refer to Drawing 1695-D-62548, 1695-D-62549, 1695-D-62550, 1695-D-62551, 1695-D-62552, 1695-D-62553, and 1695-D-62554.

1.03 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM A139/A139M-16	Electric-Fusion (Arc) - Welded Steel Pipe (NPS 4 and Over)
2.	ASTM C150/C150M-17	Portland Cement
3.	ASTM A283/A283M-18	Low and Intermediate Tensile Strength Carbon Steel Plates
4.	ASTM A1011/A1011M-15	Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Allow and High-Strength Low-Allow with Improved Formability
5.	ASTM A1018/A1018M-16a5	Steel, Sheet and Strip, Heavy Thickness Coils, Hot Rolled, Carbon, Structural, High-Strength Low-Alloy, Columbium or Vanadium, and High-Strength Low-Alloy with Improved Formability
6.	ASTM E165/E165M-12	Liquid Penetrant Examination for General Industry
7.	ASTM E709/E709M-15	Standard Guide for Magnetic Particle Testing

B. American Welding Society, Inc. (AWS)

1.	AWS D1.1/D1.1M-15	Structural Welding Code – Steel

C. American Water Works Association (AWWA)

1.	AWWA C200-12	Steel Water Pipe - 6 Inch (150mm) and Larger
2.	AWWA C205-12	Cement-Mortar Protective Lining and Coating for Steel Water Pipe - 4 Inch (100mm) and Larger - Shop Applied
3.	AWWA C206-17	Field Welding of Steel Water Pipe
4.	AWWA C208-12	Dimensions for Fabricated Steel Water Pipe Fittings
5.	AWWA C227-11	Bolted, Split-Sleeve Restrained and Non-restrained Couplings for Plain-End Pipe
6.	AWWA C604-11	Installation of Buried Steel Water Pipe 4 Inch (100 mm) and Larger

7.	AWWA M11-04	Steel Pipe: A Guide for Design and Installation, Fourth Edition

1.04 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 33 11 12-1, Shop Drawings:

1. Show pipe and fitting fabrication details.
2. Show exact dimensions of joints, position, marking of pipe sections and proposed joint restraint.
3. Show exact dimensions of welding lead holes and repair method.
4. Show details of factory epoxy lining of joint cement mortar hold back area if alternate epoxy joint recess lining is proposed.
5. Include centerline and invert stationing and elevations at horizontal and vertical changes in alignment, and subgrade elevation for each pipe segment and fitting.
6. Include pipe thickness or pressure class of pipe segment or fitting.

C. RSN 33 11 12-2, Ventilation Plan:

1. Method for ventilating inside of pipe when placing mortar lining for joints.

D. RSN 33 11 12-3, Field Welder Qualifications

1. Welder certifications

E. RSN 33 11 12-4, Filling and Testing Plan:

1. Proposed rate, time, and procedure for:
   a. Cleaning.
   b. Filling.
   c. Field and pressure testing.
   d. Draining pipeline.
2. Method for disposing of water drained from pipeline to enable repair of leaks.
3. Leak testing equipment and data sampling plan.

F. RSN 33 11 12-5, Flotation Prevention Plan:

1. Show location and describe method of preventing pipe from floating.
2. Provide supporting calculations.

G. RSN 33 11 12-6, Pipe Deflection Measurements.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Ship pipe after receiving COR approval as per Section 2.03 – Source Quality Assurance.

B. Prevent damage to pipe and fittings during loading, transporting, unloading, storing, and laying.

C. Transport metallic pipe and fittings on padded bolsters curved to fit outside of pipes. Use heavy padding under ties.

D. Tightly close open ends of shop-applied, cement-mortar-lined pipe with plastic wrap for protection of cement-mortar lining during shipment:

1. Plastic Wrap:

a. At least 2 thicknesses of 6-mil sheet polyethylene plastic.

b. Remain on pipe until installation.

E. Support and store pipe above ground surface. Do not allow bells and spigots to contact each other or the ground.

F. Government will inspect pipe once it is delivered.

G. Replace or repair, as approved by COR, any pipe that is damaged during shipment, storage, or installation at Contractor's expense. Include linings and coatings.

1.06 QUALIFICATIONS

A. Pipe manufacturers shall be certified in accordance with ISO:9001 or similar nationally certified program approved by COR.

B. Line pipe and specials manufacturers:

1. 10-years successful experience producing products as specified.

2. Provide references for at least 3 completed projects with similar pipe diameters (within 12-inches) and pressures.

C. Pipe Layers: Skilled and experienced in laying pipe with joints as needed for each material type of pipe used. Minimum 2-years installing pipe with similar pipe diameters (within 12-inches), pipe material, and pressures:

1. In the absence of specific pipe material experience, substitution of onsite pipe manufacturer training and quality control is allowed:

a. Provide manufacturer's representative certified in pipeline installation, including all applicable joints, for training pipe layers and pipe foreman prior to installation.

b. In lieu of specific pipe material installation experience, pipe layers and pipe foreman shall still have experience with installing similar pipe diameters and pressures.

D. Qualify welding procedures and welders in accordance with code under which welding is specified.

PART 2 PRODUCTS

2.01 STEEL LINE PIPE

A. Pipe:

1. Electric fusion (arc) welded helical-seam steel pipe: ASTM A139, Grade C, D, or E.
2. Fabricated in accordance with AWWA C200:
   a. Except:
      1) Steel plate: ASTM A283, Grade C or D.
      2) Steel Sheet or coil: ASTM A1011, designation SS, Grade 40, 45, or 50; or ASTM A1018, designation SS, Grade 40 or designation HSLAS-F, Grade 50.

B. Inside diameter measured to inside of lining shall not be less than nominal diameter shown on drawings.

C. Minimum Steel Wall Thickness:

1. Greater thickness from the following requirements:
   a. Steel Wall Thickness for Handling:
      1) Equal to inside diameter (inches) of pipe steel shell divided by 240 for mortar lined and flexible coated steel pipe.
   b. Steel Wall Thickness for Internal Pressure:
      1) Hoop stress of pipe shell. Not to exceed:
         a) 50 percent yield strength as defined by steel grade of steel using the Barlow formula. With pressure defined by Hydraulic Grade Line as shown on pipeline plan and profile drawings.
         b) 75 percent yield strength as defined by grade of steel using the Barlow formula. With pressure defined by pipe head class as shown on pipeline plan and profile drawings.
      2) Barlow formula as defined by AWWA M11 Equation 4-1.

D. Steel shall be fully killed and conform to fine grain practice.

E. Rubber gaskets: In accordance with AWWA C200.

F. Materials and chemicals that may come into contact with drinking water shall be certified by NSF 61.

2.02 FITTINGS

A. Steel: ASTM A283, Grade C or D or ASTM A1011, designation SS, Grade 40, 45, or 50; or ASTM A1018, designation SS, Grade 40 or designation HSLAS-F, Grade 50.

B. Minimum steel wall thickness and fitting dimensions: In accordance with AWWA C200, AWWA C208 and AWWA M-11.

C. Welding: In accordance with AWS D1.1.

D. Lifting eyes and other handling devices: Made part of fitting before lining and coating are applied.

E. Bolts and Nuts: In accordance with ASME B18.2.1 and FS FF-N-836.

F. Coating and lining in accordance with Section 09 96 20 – Coatings.

G. Cement-Mortar Lining for Specials and Bends Larger than 24-inches:

1. In accordance with AWWA C205.
2. Apply lining with a resultant surface finish, including joints, equal to finish of adjacent pipe.

H. Joints between fittings and ductile iron pipe:

1. Rubber gasket joints.
2. Joint dimensions and tolerances: Same as pipe manufacturer's joint design.

I. Miter Bends: Fabricate steel bends in accordance with AWWA C208, AWWA M-11 and as shown on drawings.

J. Pipe Couplings:

1. Coupling types as shown on drawings.
2. Suitable for line pipe materials used.
3. Suitable for cold water pressures of pipe head classes on drawings.
4. Refer to Section 35 21 95 – Steel Manifold Piping for coupling requirements.

K. Closure Section Joints

1. Restraint requirements shown on drawings.
2. Field welded butt strap joints.

3. Bolted, Split-Sleeve Restrained and Nonrestrained Couplings for Plain – End:

   a. In accordance with AWWA C227.

   b. Refer to Section 35 21 95 – Steel Manifold Piping for coupling requirements.

L. Tees:

1. Tees for air valves, blowoffs, manholes, and temporary construction line as shown on drawings.

2. Tee length, minimum: As shown on drawings, required for blocking, or in accordance with AWWA C208 whichever is greater

2.03 LININGS AND COATINGS

A. In accordance with Section 09 96 20 – Coatings.

B. Do not field apply mortar lining without OGR approval.

2.04 CONCRETE

A. Concrete in Thrust Blocks and Collars: Refer to Section 03 30 00 – Cast-in-Place Concrete.

2.05 FLANGES:

A. In accordance with AWWA C207.

B. Class: Working pressure meets or exceeds head class of attached pipe

2.06 FLANGE GASKETS:

A. BLUE-GARD Style 3000 manufactured by Garlock Sealing Technologies, 1666 Division Street, Palmyra NY 14522, or equal, having the following essential characteristics:

1. Sized in accordance with AWWA C207, Table 1.

2. For AWWA C207 flanged joints.

3. Compressed, Non-Asbestos (CNA) Gasketing with Aramid Fibers and a NGR Binder.

4. For potable cold-water service.

2.07 PIPE ACCESSORIES

A. Sectionalizing Valve, Air Valve, and Blowoff Assemblies: In accordance with Section 35 22 14 – Valves and Equipment.

B. Buried Manhole Assemblies:

1. In accordance with Section 03 48 00 – Precast Concrete Structures.

2. Cement-mortar lining: AWWA C205, in accordance with shop applied mortar for lining of specials, except:
    a. Reinforcement required for pipe larger than 24-inch diameter.
    b. Cement: ASTM C150, Type V.
3. Required at locations shown on drawings.

C. Warning Tape:
1. Polyethylene tape.
2. Thickness: 4 mils.
3. Magnetically detectable.
4. Width, minimum: 6-inches.
5. Color: Blue with silver lettering.
6. Legend repeated at least every 3-feet: "CAUTION: WATERLINE BURIED BELOW - FOR MORE INFORMATION CONTACT THE NAVAJO TRIBAL UTILITY AUTHORITY".

2.08 CONTRACTOR SOURCE QUALITY TESTING

A. Pipe and Fittings:
1. Hydrostatic Test:
    a. Perform shop hydrostatic test on pipe which stresses steel to 75 percent of minimum yield point of the steel.
    b. Hold pressure long enough to allow thorough inspection of welded joints.
    c. Repair leaks by rewelding and retesting joints.
    d. Test sections prior to forming bell and spigot joints.
    e. Repair defects and retest section before applying lining and coating.
    f. Government may witness hydrostatic testing and calibration of pressure gauges. Notify Government 30 days prior to performing hydrostatic test.
2. Fittings fabricated from tested steel pipe do not require hydrostatic testing if welds are tested.
    a. Weld test: Dye penetrant test in accordance with ASTM E165 or Magnetic Particle in accordance with ASTM E709.
    b. Weld defect repair. Chip, flame gouge, or grind to sound metal; reweld; and test repaired weld.

B. Joint Testing: Assemble one joint for each pipe diameter to check bell and spigot to check fit prior to coating or lining.

PART 3 EXECUTION

3.01 GENERAL

A. Perform cleaning, filling, and testing after backfill has been placed to finished grade or as approved by OGR.

B. Responsible Surveyor licensed in State of New Mexico or a certified Construction Surveyor, as defined in and in accordance with Section 01 71 20 – Surveying, shall be onsite during excavation and pipelaying.

C. Protect pipe from contamination, damage, and elements during storage on-site by covering with tarp as directed by OGR.

D. Install pipe based on these specifications or pipe manufacturer's recommendation, whichever is more stringent.

E. Isolate dissimilar metals.

3.02 INSTALLATION

A. Keep openings to installed pipe closed with watertight inflatable plugs during work stoppage, including end of work day, breaks, and work delays.

B. If pipe is flooded during construction, clear floodwater by draining and flushing with water, or other approved method, until pipe is clean.

C. Lubricant:

1. Keep clean.
2. Apply with dedicated, clean applicator brushes.
3. Apply lubricant as specified by pipe manufacturer:
    a. Use only lubricant supplied by pipe manufacturer.
    b. Lubricate spigot end of pipe as recommended by manufacturer.
    c. Do not lubricate either gasket or gasket groove in bells where gaskets are field installed.
    d. Lubricate factory-installed, non-removable-type gaskets as recommended by pipe manufacturer.

D. Coat buried manholes with cement mortar as shown on drawings.

E. Install warning tape over center of pipe at least 18-inches below ground and at least 18-inches above pipe.

3.03 LAYING PIPE

A. Excavate pipe trench in accordance with Section 31 23 22 – Pipe Trench Earthwork.

1. Grade pipe trenches to provide uniform slope along bottom of pipe.
2. At joints involving bells or collars, provide holes at joint of ample size to prevent bells or collars from coming in contact with subgrade and to allow for bedding placement.

B. Keep pipe trenches free of water during pipelaying operations.

C. Lower pipe into trench and place pipe in position such that no soil gets inside pipe and pipe is not damaged.

D. Install pipe to line and grade as shown on drawings.

E. On grades exceeding 10 percent, lay pipe uphill.

F. When pipelaying is not in progress, keep ends of pipelines closed with watertight inflatable plugs.

G. Joining Pipe:

1. In accordance with AWWA C604.
2. Verify inside joint gap meets manufacturer's tolerances as work progresses.

H. Joints for Pipe with Welded Joints:

1. In accordance with AWWA C206. Welded joints after backfill not allowed.
2. Perform dye penetrant test on welds in accordance with ASTM E165.
3. Welded Butt Joints:
   a. As required for pipe on steep slopes as shown on drawings.
4. Double Lap Weld Joints:
   a. Provide joint with full fillet welds.
   b. Air test double lap welds in accordance with AWWA C206.
   c. Required on thrust restrained joints greater than 30 degree bend deflection angles, unless butt welded joints are required.
5. Single Lap Weld Joints:
   a. May be used on thrust restrained joints with 30 degree bend deflection angles or less.
   b. Road and wash crossings unless butt or double-lap welded joints are required.

I. Connections at Thrust Blocks and Structures: As shown on drawings

J. Closure Sections:

1. Where necessary as determined by Contractor and subject to approval of OGR.
2. Ambient temperature when closure section is welded, maximum: 60 degrees F.

3. Assemble push-on or mechanical joints in accordance with applicable AWWA standard for pipe type and manufacturer's recommendations.
4. If adjustment of the position of length of pipe is required after installation, remove and relay length of pipe as new pipe.
5. Maintain pipe firmly in final position.
6. Placing Elastomeric Gasket:
    a. Clean gasket, bell, especially the groove, and spigot with rag, brush, or paper towel to remove any dirt or foreign material.
    b. Use only gaskets which are designed for and supplied with pipe.
    c. Lubricate pipe as recommended by manufacturer.
    d. Spigot Groove Method:
        1) Follow manufacturer's recommended practices for gasket installation.
        2) After placing elastomeric gasket in spigot groove, equalize elastomeric gasket cross section by inserting tool such as a large screwdriver under elastomeric gasket and moving it around periphery of pipe spigot.
    e. Coupling Groove Method: Follow manufacturer's recommended practices for gasket installation. Typically, factory installed.
7. Fit pipe unit's together, spigot to bell or coupling, and draw joints together so that bells or couplings and spigots are fully engaged with uniform contact to gasket.
8. Do not swing or "stab" joint and do not suspend pipe and swing into bell or coupling.
9. Fit pipe units together in a manner to avoid twisting or otherwise displacing or damaging elastomeric gasket.
10. After joining pipe units, insert feeler gauge between pipe bell and pipe spigot, move it around periphery of pipe to determine that position of elastomeric gasket is correct and there are no fish-mouth problems. If adjustment of position of a length of pipe is required after installation, remove and relay length of pipe as new pipe.

K. Changes in Alignment and Grade:
1. Make changes in alignment and grade with miter bends as shown on drawings.
2. Make other minor changes in alignment and grade by providing small deflections between adjacent pipe:
    a. Do not exceed manufacturer's deflection tolerances.
    b. Pipelaying diagrams shall only deflect joints no more than 50-percent of manufacturer's maximum deflection tolerances.

3. Provide restrained joint lengths upstream and downstream of bends or thrust blocks as shown on drawings. Ductile iron fittings with restrained joint lengths as limited by drawings may be used if Contractor submits revised plan and profile drawings for approval.

4. Do not encase rubber gasket joints in concrete.

5. Refer to individual pipe Sections for additional requirements. If requirements conflict, use most stringent requirement.

L. After pipelaying and joining operations are completed, clean inside of pipe and remove debris.

M. Schedule work so that at no time pipe remains in trench more than 7-days before backfill is placed to original ground surface or to other specified backfill limits shown on drawings.

3.04 THRUST RESTRAINT

A. Restrained joint couplings or concrete blocking and collars as shown on drawings. Refer to Section 03 30 00 – Cast-in-Place Concrete and in accordance with applicable sections listed in 2.01.A.

B. Plan, profile, and restraint system shown on drawings require modification if alternate methods are used.

C. Alternate methods of restraint may not be used unless submitted and approved by COR.

3.05 TOLERANCES

A. Lay pipe to lines and grades shown on drawings or established by OGR to following tolerances:

1. Total departure from established alignment and grade, maximum: 1-inch.

3.06 ADJUSTMENTS IN ALIGNMENT AND GRADE FOR PULLED JOINTS

A. Per approved shop drawings.

B. Make small minor changes in alignment and grade by providing small deflections between adjacent pipe sections as follows:

1. Pulling rubber gasket bell-and-spigot joint:

a. A total 1-inch joint deflection may be permitted by reducing normal 1/2-inch inside mortar space to 1/4-inch and pulling opposite side of pipe 3/4-inch from normal closure.

b. Provide unsymmetrical closure at welded lap joint up to maximum pullout of 1-inch.

2. Maximum deflection angle between adjacent pipe sections: Manufacture's recommendations but shall not exceed 5 degrees.

C. Lay ends of each section of steel line pipe on theoretical centerline of curve and to grade shown on drawings within laying tolerances prescribed above.

3.07 JOINT LINING AND COATING

A. Field joints for pipe with shop applied cement-mortar lining.

1. Line field joints with cement-mortar in accordance with AWWA C205.
2. Coat exterior joints as specified in Section 09 96 20 – Coatings.
3. Do not backfill field joints until coating is approved by OGR.

B. Apply shrink sleeves to field joints in accordance with Section 09 96 20 – Coatings.

3.08 BACKFILL

A. Keep internal supports in place until embedment has been placed and compacted above bottom of pipe to minimum height of 0.7 times diameter.

3.09 PIPE DIAMETER DEFLECTION

A. Allowable short-term vertical pipe diameter deflection after backfilling is complete.

1. Decrease, maximum: 2 percent of nominal pipe diameter.
2. Elongation, maximum: 3 percent of nominal pipe diameter as measured when backfill reaches pipe crown.
3. Short term is within 2 weeks after backfilling is complete.

B. Allowable long-term vertical pipe diameter deflection, at end of warranty period:

1. Decrease, maximum: 3 percent of nominal pipe diameter.

END OF SECTION

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SECTION 35 21 95
STEEL MANIFOLD PIPING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Steel Manifold Piping:

1. Payment: Lump sum price offered in Price Schedules.
    a. Includes cost of pipe and fittings within the turnout buildings.
    b. Includes cost of pipe and fittings for yard pipe located outside of the turnout buildings.

1.02 DEFINITIONS

A. Steel Manifold Piping:

1. Details of steel manifolds are shown on drawings 1695-D-62558 through 1695-D-62568.

1.03 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)

1.	ASME B1.20.1-2013(R2018)	Pipe Threads, General Purpose (Inch)
2.	ASME B16.9-2024	Factory Made Wrought Buttwelding Fittings
3.	ASME B16.11-2021	Forged Fittings, Socket-Welding and Threaded

B. ASTM International (ASTM)

1.	ASTM A36/A36M-19	Carbon Structural Steel
2.	ASTM A53/A53M-24	Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
3.	ASTM A105/A105M-24	Carbon Steel Forgings for Piping Applications
4.	ASTM A106/A106M-26	Seamless Carbon Steel Pipe for High-Temperature Service
5.	ASTM A139/A139M-22	Electric-Fusion (Arc)-Welded Steel Pipe (NPS 4 and Over)
6.	ASTM A194/A194M-24	Carbon Steel, Alloy Steel, and Stainless-Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both

7.	ASTM A234/A234M-25	Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
8.	ASTM A420/A420M-25	Piping Fittings of Wrought Carbon Steel and Alloy Steel for Low Temperature Service
9.	ASTM A1011/A1011M-25	Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
10.	ASTM A1018/A1018M-25	Steel, Sheet and Strip, Heavy Thickness Coils, Hot Rolled, Carbon, Commercial, Drawing, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
11.	ASTM E165/E165M-23	Liquid Penetrant Examination for General Industry
12.	ASTM E709-21	Magnetic Particle Testing

C. American Welding Society (AWS)

1.	AWS D1.1/D1.1M:2025	Structural Welding Code - Steel

D. American Water Works Association (AWWA)

1.	AWWA C200-23	Steel Water Pipe - 6 In (150 mm) and Larger
2.	AWWA C206-23	Field Welding of Steel Water Pipe
3.	AWWA C207-23	Steel Pipe Flanges for Waterworks Service, Sizes 4 In. through 144 In. (100 mm Through 3,600 mm)
4.	AWWA C208-22	Dimensions for Fabricated Steel Water Pipe Fittings
5.	AWWA C604-24	Installation of Buried Steel Water Pipe – 4 In. (100 mm) and Larger

E. Manufacturer's Standardization Society (MSS)

1.	MSS SP-97-25	Integrally Reinforced Forged Branch Outlet Fittings: Socket Welding, Threaded, and Buttwelding Ends

F. National Sanitation Foundation International / American National Standards Institute (NSF/ANSI)

1.	NSF/ANSI/CAN 61-24	Drinking Water System Components – Health Effects
2.	NSF/ANSI/CAN 372-24	Drinking Water System Components – Lead Content

1.04 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 35 21 95-1, Shop Drawings:

1. Checked, detailed shop drawings including dimensions and tolerances, material data, joint details, and welding requirements:
    a. Provide checked box on each drawing indicating date drawing was checked and by whom. Drawings must be stamped by a PE (Professional Engineer).
    b. When revised drawings and data are resubmitted, clearly identify changes from previous submittals.

C. RSN 35 21 95-2, Commercial Products Data:

1. Items in Part 2 Products.
2. Manufacturer's technical product data, including the applicable sizes, standards, material, and pressure rating.
3. ***Data or calculations that demonstrate differential settlement is accommodated within allowable space for pipe couplings above 24-inch.***

D. RSN 35 21 95-3, Final Data:

1. Weld nondestructive testing (NDT) reports including retests after weld repair.
2. Results of air and soap tests on field welded double lap and butt strap joints.
3. Hydrostatic test reports, including time-pressure record.
4. Affidavit of compliance to AWWA C200 and C206 as applicable.

E. RSN 35 21 95-4, Final Drawings:

1. As-Built drawings
    a. Clearly mark drawings 1695-D-62558 through 1695-D-62568 to show all field modifications of the final installation.
    b. Submit drawings in accordance with Section 01 33 00 – Submittals article 1.04 paragraph C.6 and Section 01 78 30 –Project Record Documents.

1.05 QUALIFICATIONS

A. Pipe Installers: Installed similar manifold on 3 projects within the last 5 years.

B. Steel Pipe Welders: Welders and Welding Operators to be qualified in accordance with AWS D1.1.

1.06 DELIVERY, STORAGE, AND HANDLING

A. During loading, transporting, unloading, storing, and laying, prevent damage to steel manifold piping linings and coating. Transport and handle in accordance with fabricator's instructions:

1. Provide padded bolsters curved to fit under outside of pipe.
2. Provide heavy padding under ties during transportation and storage.

B. Maintain circularity of pipe sections during coating, lining, transporting, and installing operations:

1. Provide stulls or adjustable spiders.
2. Measured deviation between maximum and minimum diameters at stulls or spiders shall not exceed 0.5 percent of nominal pipe diameter.

C. Do not store directly on the ground.

D. Store so that lifting straps can be passed beneath pipe sections.

E. Provide space to allow personnel to move between pipe sections.

F. Cover ends with tarps and tie down securely.

G. Repair damage to lining and coating as directed if, in the opinion of COR, a satisfactory repair can be made; otherwise, replace damaged section at expense of Contractor and in accordance with Section 09 96 20 – Coatings.

PART 2 PRODUCTS

2.01 GENERAL REQUIREMENTS

A. Piping and equipment materials:

1. ANSI/NSF/CAN 61 certified.
2. ANSI/NSF/CAN 372 certified lead-free.
3. Suitable for potable water.
4. Materials suitable for free chlorine residual exposure.
    a. Typical free chlorine residual concentration: Maximum 4 milligrams per liter (mg/l).

2.02 EQUIPMENT

A. Steel plate: ASTM A36, ASTM A1011, SS, Grade 36, 40, or 45; or ASTM A1018, SS, Grade 36 or 40.

B. Standard and Schedule Steel Pipe:

1. ASTM A53, Type E or S, Grade B:

a. Pipe 2-inches in diameter: Fusion Bonded Epoxy.

1) Acceptable alternative ASTM A106.

b. Pipe 3-inches and larger in diameter: In accordance with Section 09 96 20 – Coatings.

C. Electric–Fusion– (arc–) Welded, Spiral–Seam Steel Pipe: ASTM A139, Grade C, D, or E.

D. Corrosion Collar:

1. Steel plate: ASTM A36, ASTM A1011, Grade 36, 40, or 45; or ASTM A1018, Grade 36 or 40.

E. Flanges:

1. Steel: Slip-on type, flat faced, AWWA C207.
2. Flat face with finish suitable for gasket: Provide spiral or concentric serrated grooved flange face gasket surface finishes that are suitable for flat gasket material provided and for operating pressures.
3. Do not remove machine marks.

F. Gaskets:

1. Suitable for cold water service.
2. Do not reuse compressed gaskets.
3. Use Multi-Swell Style 3760 manufactured by Garlock, www.garlock.com; or equal, having the following essential characteristics:

a. Dimensional requirements of AWWA C207 as shown on drawings.

b. Flat, full face type.

c. Minimum 300 psig cold water pressure rating.

d. Minimum 1/16-inch thick.

G. Flange bolts, Nuts, and Washers:

1. AWWA C207 for AWWA flanges.
2. Bolts: ASTM A193/A193M, Grade B7.
3. Nuts:

a. Greater than 1-inch diameter: ASTM A194/A194M, Grade 2H

b. 1-inch diameter or less: ASTM A563

4. Washers: ASTM F436.

H. Pipe Flange Supports:

1. Trumbull Flange Style Adjustable Pipe Support manufactured by Trumbull Manufacturing, www.trumbull-mfg.com, or equal, having the following essential characteristics:

a. Adjustable height.

b. Material: Galvanized steel.

c. Suitable for pipe sizes as shown on the drawings.

d. Provides vertical restraint to pipe flange.

I. Threaded Fittings:

1. Threaded NPT: ASME B1.20.1.

a. Steel: ASME B1.20.1, ASME B16.11, ASTM A105 ASTM A106 and MSS SP-97.

2.

J. Butt-Welding Steel Fittings: ASME B16.9 and ASTM A234 or ASTM A420 - WPL6.

K. Concentric Steel Pipe Reducers: Designed, fabricated and dimensioned in accordance with AWWA C208.

L. Steel Pipe Miter Bend Elbows: Designed, fabricated, and dimensioned in accordance with AWWA C208.

M. Pipe couplings:

1. 24 inch and below:

a. Grooved-end: Style 77 manufactured by Victaulic, www.victaulic.com; or equal, having the following essential characteristics:

1) Flexible joint type with rolled grooves.

2) Minimum cold water working pressure rating: 300 psig.

3) Minimum angular deflection

a) 6-inch pipe diameter: 1-5' degrees

b) 8-inch pipe diameter: 0-50' degrees

c) 12-inch pipe diameter: 0-34' degrees

2. Above 24 inch:

a. Restrained couplings for dynamic joint deflection:

1) Suitable for buried service.

2) Spaced to accommodate 2-inches of differential settlement.

3) Conforms to the following options:

a) Restrained flexible type for dynamic joint deflection Style 233, manufactured by Victaulic, www.victaulic.com; or equal, having the following essential characteristics:

i. Conforms to AWWA C227.

ii. Coupling housing: Arched, 2 or more segmented, split sleeve-type which when closed compresses elastomeric O-ring gaskets beneath arches of sleeve to create radial seal around pipe ends joined inside of coupling.

iii. Minimum misalignment tolerance (lateral offset between pipe ends): 0.125-inch.

iv. Minimum allowable angular deflection for 48-inch: 1°7'.

N. Anchor bolts: In accordance with Section 03 20 60 – Concrete Anchors.

O. Flange insulation:

1. Flange isolating gasket kit complete with the "Trojan" type of gasket manufactured by Advanced Products and Systems, Inc., www.apsonline.com, or equal with the following essential characteristics:

   a. Provide full-faced gaskets as shown on drawings. Outside diameter of gasket is same as outside diameter of flange.

   b. Provide retainer with integral Nitrile sealing element ("Trojan" type).

   c. Retainer material to be suitable for cold water service as recommended by gasket manufacturer.

   d. One insulating sleeve and 2 insulating washers for each bolt. Material suitable for cold water service as recommended by gasket manufacturer.

   e. Asbestos gaskets are not acceptable.

   f. Prevent damage to gasket, gasket supplier to provide required bolt torques, recommended bolt tightening sequence, and flange face finish for insulating flanged joints.

   g. Suitable for 300 psig cold water service.

2.03 FABRICATION

A. Fabricate steel manifold piping in accordance with these specifications, as indicated on drawings, and in accordance with AWWA C200.

B. Longitudinal, girth, and spiral joints, other than field welds:

1. Double-welded butt joints with complete penetration.
2. Stagger longitudinal joints.
3. Longitudinal, girth, and spiral joints shall not intersect at outlet connections.

C. Cut plates accurately to size and shape.

D. Bend or roll plates to true circular with curvature continuous from edges of plates.

E. Properly form edges to be joined by welding to suit type of welding and to allow complete penetration.

F. Perform welding with a process that protects molten metal from atmosphere:

1. Where practicable, use automatic machines.
2. Where weld metal is deposited in successive layers, clean each layer thoroughly before subsequent layer is deposited.
3. Take particular care in aligning and separating edges of plates to be joined by butt welding so that complete penetration and fusion of welds will be ensured.
4. After welding is complete, remove weld spatter.

G. Protect work and operator from wind, rain, and snow during welding operations:

1. Welding is not permitted on wet surfaces or when temperature of steel is lower than 0 degrees F.
2. Apply heat to welding areas as needed during field installation. At pipe surface temperatures between 0 and 32 degrees F, heat surface of areas within 3-inches of the point where a weld is to be started to at least 60 degrees F.

H. Ends to be joined by a pipe coupling: Exterior surface of pipe end to be connected with sleeve-type coupling shall be free from weld reinforcement (bead), seams, scars, dents, flats, or other defects likely to interfere with joint-sealing process in accordance with coupling manufacturer instructions.

I. Flanges and flanged connections:

1. During attachment of flanges to associated steel manifold pipe, take adequate steps to ensure that face of flanges remains flat and perpendicular to centerline of attached piping.
2. Remove by machining, initial, residual, or latent irregularities or warping that remains in face of flange after its attachment to steel manifold piping.
3. After machining, flange thickness shall not be reduced below minimum thickness specified in AWWA C207, or as shown on drawings.
4. Weld pipe flanges to the piping so that bolt holes straddle vertical centerline when in the installed position.

5. Provide pipe flanges that connect to valve body flanges with same number of bolt holes, bolt pattern, bolt circle diameter, and face type, flat or raised, as valves to which they are connected.

J. Continuously weld joints of flange supports and pipe supports.

2.04 CONTRACTOR SOURCE QUALITY TESTING

A. Flange Weld Testing:

1. Test flange welds by magnetic particle or liquid penetrant method in accordance with AWS D1.1.
2. Repair defects and retest weld.

B. Hydrostatic Pressure Testing:

1. Straight sections of steel piping manufactured and hydrostatically tested in accordance with ASTM A53 or ASTM A139 need not be retested.
2. Notify Government a minimum of 30-days before test.
3. Government may witness hydrostatic test and calibration of pressure gauges.
4. Use 2 or more pressure gauges with one being a recording type:
   a. Upper limit of pressure gauges shall be between 1.5 and 3 times required test pressure. Calibrate pressure gauges with a deadweight tester.
   b. Calibrate at least 1 gauge, which is used for acceptance of test, within 30-days prior to use.
   c. Make a time-pressure record of hydrostatic pressure test.
5. Furnish temporary supports, bulkheads, air vents, dead weight tester, and other equipment necessary to perform hydrostatic tests.
6. If flange is attached to piping section after piping section has been hydrostatically tested, piping section will not require retesting.
7. Water temperature for tests: Greater than 45 degrees F.
8. Vent air from test sections.
9. Fill each section with water and uniformly increase pressure until required test pressure is reached:
   a. Hold test pressure for a minimum of 15-minutes.
   b. Apply and release test pressure 3 successive times.
   c. After each application, hold pressure to 2/3 of required pressure until welded joints and seams can be examined.
   d. Repair defects. After defects are repaired repeat complete test procedure.
10. After hydrostatic testing is completed, cut sections for handling purposes on planes normal to pipe axis.

PART 3 EXECUTION

3.01 PREPARATION

A. Locations of existing lateral turnout tie-ins are approximate, confirm locations prior to fabrication of steel manifold piping.

B. Construction of cPWS distribution pipes is ongoing and may be constructed at some sites. Coordinate with COR to verify consecutive distribution entry points.

3.02 INSTALLATION

A. Install steel manifold piping to line and grade as shown on drawings and in accordance with AWWA C604:

B. Prevent steel pipe from being displaced or overstressed during concrete and backfill placement.

C. Contact between pipe and rebar will not be allowed.

D. Install pipe spools with flanged connections so that bolt holes straddle vertical centerline when in installed position.

E. Dielectrically insulate between dissimilar metals. Provide insulating type unions at threaded valve, pipe or fitting connections between dissimilar metals.

F. Install pipe couplings, gaskets, and other appurtenances in accordance with manufacturer instructions:

1. Maximum allowable misalignment (lateral offset) between pipe ends at split sleeve type couplings: 0.125-inch:
    a. Government may witness installation of pipe couplings.
    b. If misalignment exceeds 0.125-inch, piping shall be repaired at the expense of the Contractor to correct misalignment. Piping repair method subject to COR approval.
    c. Do not deflect pipe between coupling pairs at couplings for dynamic joint deflection to correct misalignment.

G. Install anchor bolts in accordance with manufacturer instructions.

H. Field joints:

1. Field joints shown on drawings are required to facilitate lining or installation of steel piping.
2. Allowable field joints for steel piping:
    a. AWWA C207, AWWA C228 flanged joints per ASME PCC-1 respectively:

1) Flanges shall be suitable for working pressure of adjacent pipe.

2) Flanges shall not be permitted underneath buildings, , or in concrete encasements.

b. Full-penetration butt welds.

c. Double welded lap joints, or double welded butt strap joints.

d. Threaded couplings for pipe less than 4-inches in diameter.

3. Welded field joints shall be in accordance with AWWA C206 or AWWA C231.

4. Prepare field welded surfaces and recoat in accordance with Section 09 96 20 – Coatings.

I. Steel manifold piping earthwork: In accordance with Section 31 23 22 – Pipe Trench Earthwork.

3.03 CONTRACTOR FIELD QUALITY TESTING

A. Nondestructive Testing of Field Welds of Steel Pipe:

1. Test all welds in accordance with AWWA C206 and this specification.

2. Complete inspections and tests as work progresses in presence of COR and in accordance with AWS D1.1.

3. When welds are found to be defective by the standards of AWS, repair welds in accordance with AWS D1.1 and reexamine welds to ensure the adequacy of repairs.

4. Visual Inspection: Visually inspect welds following procedures, techniques, and standards in accordance with AWS D1.1.

5. Testing of full penetration joints:

a. Ultrasonic methods in accordance with AWS D1.1.

6. Testing of partial penetration joint field welds and fillet field welds, including testing of double welded butt strap joints or double welded lap joints:

a. Magnetic particle testing: ASTM E709.

b. Dye penetrant: ASTM E165.

c. Standards of acceptance for both types: AWS D1.1.

d. Caulking or peening of defective welds is not permitted.

e. Repair and retest unsatisfactory welds.

7. Inspect all double lap welded and butt strap joints by air and soap test:

a. AWWA C206.

b. Test the joints by shop drilling and tapping for a 1/8-inch or 1/4-inch national pipe thread in the lap end of the pipe.

c. Apply 40 psig of air into the connection between the two fillet welds and maintain for a period of 5-minutes.

d. Paint the welds with a soap solution.

e. Mark any leaks indicated by the escaping air bubbles.

f. Repair welds at leak locations in accordance with AWS D1.1 and AWWA C206 and retest.

g. Close the threaded openings with pipe plugs.

h. Seal weld the pipe plugs and grind the weld smooth.

i. Submit test results to COR within one week after testing.

j. Prepare field welded surfaces and recoat in accordance with Section 09 96 20 – Coatings.

3.04 LINING AND COATING

A. Refer to Section 09 96 20 – Coatings.

END OF SECTION

SECTION 35 22 14
VALVES AND EQUIPMENT

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Valves and Equipment:

1. Payment: Lump sum prices offered in Price Schedules for various types and sizes of valves and equipment provided as shown on drawings.

1.02 DEFINITIONS

A. Details of valves and equipment are shown on drawings: 1695-D-62548 through 1695-D-62552 and 1695-D-62558 through 1695-D-62568.

1.03 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)

1.	ASME B16.5-2025	Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch

B. ASTM International (ASTM)

1.	ASTM A216/A216M-21	Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service
2.	ASTM A536-24	Ductile Iron Castings

C. American Society of Safety Engineers (ASSE)

1.	ASSE 1063-21	Air Valve and Vent Inflow Preventer

D. American Water Works Association (AWWA)

1.	AWWA C207-23	Steel Pipe Flanges for Waterworks Service, Sizes 4 In. through 144 In.
2.	AWWA C504-23	Rubber-Seated Butterfly Valves
3.	AWWA C507-23	Ball Valves, 6 In. Through 60 In. (150 mm Through 1,500 mm)
4.	AWWA C508-25	Swing-Check Valves for Waterworks Service, 2-In. through 48-In. (50-mm Through 1200-mm) NPS
5.	AWWA C509-23	Resilient-Seated Gate Valves for Water Supply Service

6.	AWWA C512-25	Air-Release, Air/Vacuum, and Combination Air Valves for Water and Wastewater Service
7.	AWWA C514-21	Air Valve and Vent Inflow Preventer Assemblies for Potable Water Distribution System and Storage Facilities
8.	AWWA C515-20	Reduced-Wall, Resilient-Seated Gate Valves for Water Supply Service
9.	AWWA C530-22	Pilot Operated Control Valves

E. Manufacturers Standardization Society of the Valves and Fittings Industry (MSS)

1.	MSS SP-110-2010	Ball Valves Threaded, Socket–Welding, Solder Joint, Grooved, and Flared Ends

F. National Sanitation Foundation International / American National Standards Institute (NSF/ANSI)

1.	NSF/ANSI/CAN 61-24	Drinking Water System Components – Health Effects
2.	NSF/ANSI/CAN 372-24	Drinking Water System Components – Lead Content

1.04 SUBMITTALS

A. Submit the following in accordance with the requirements of Sections 01 33 00 – Submittals.

B. RSN 35 22 14–1, Commercial Products Data:

1. For items in Part 2 Products.
2. Manufacturer's technical product data, including the applicable size, standards, bill of materials, and pressure rating:
    a. For pilot-operated control valves, include the following for each type and size of valve:
        1) Parts identification lists.
        2) Lists of special tools and accessories.
        3) Schematics showing pilot-operated control system, connections and manufacturer recommended componentry:
        4) Instructions for installing, adjusting, operating, lubricating, and maintaining equipment.
        5) Cavitation chart showing flow rate, differential pressure, valve characteristic curve (Cv versus percent open), and system velocity.

C. RSN 35 22 14-2, Field Installation Procedures.

1. For items listed in Part 2 Products.
2. Pilot-operated control valves: Include circuit schematic of pilot-operating system.

D. RSN 35 22 14–3, Shop Testing Certifications, Reports, and Data:

1. Hydrostatic pressure testing certifications and pressures used in testing valves:
   a. AWWA gate valves tested in accordance with AWWA C509, or AWWA C515.
   b. Pilot-operated control valves tested in accordance with AWWA C530.
   c. Silent check valves tested in accordance with AWWA C508.
2. Affidavit of compliance to applicable standards:
   a. AWWA gate valves.
   b. AWWA silent check valves.
   c. Pilot-operated control valves.
3. Include pertinent observations and notable occurrences during testing.

E. RSN 35 22 14–4, Service Manuals and Final Data:

1. Items in Part 2 Products.
2. Complete parts identification lists including clearly identified recommended spare parts, lists of special tools and accessories, detailed instructions for installing, adjusting, operating, lubricating, and maintaining the equipment.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Take precautions during loading, transportation, unloading, storage, and laying, to prevent damage to the valves and equipment:

1. Repair damage to the valves and equipment as directed if, in the opinion of COR, a satisfactory repair can be made; otherwise replace damaged valves and equipment.
2. Ship, handle, store, and position according to manufacturer's instructions.
3. Cover valves until installed.

1.06 LOCATION DATA

A. Refer to table as shown on drawing 1695-D-62558 for flow control valve sizes.

PART 2 PRODUCTS

2.01 GENERAL REQUIREMENTS

A. Valves and equipment materials:

1. ANSI/NSF/CAN 61 certified.
2. ANSI/NSF/CAN 372 certified lead-free.
3. Suitable for potable water.

B. Valve Body Flange Faces and Flanges:

1. Valve manufacturers are responsible for providing the correct flange face gasket surface finish.
2. Provide spiral or concentric serrated grooved flange face gasket surface finishes that are suitable for the gasket material provided and for the operating pressures.
3. Valve flange bolt patterns to match pipe flanges.

C. Furnish valves and actuators of same type and size from same manufacturer.

D. Spare parts:

1. Provide one extra set of seats for each size and type of gate valve.
2. parts for each size and type of air valve including:
    1) Seats, lever, and gasket.
    2) Additional parts recommended by manufacturer.
    3) Box and clearly label box: Include list of spare parts inside box.
3. Store spare parts in location as directed by COR.

2.02 VALVES AND OPERATOR

A. Pilot Operated Control Valves with Solenoid Shut-off:

1. General:
    a. AWWA C530.
    b. Full, globe style.
    c. Anti-cavitation trim as recommended by manufacturer.
    d. During valve operation, water vented from upper pressure chamber of main valve (when main valve is opening) shall be directed into pilot piping so that it discharges into downstream side of main control valve.
    e. Diaphragm operated.
    f. Pressure gauges integral to the main control valve to measure upstream and downstream pressures.

g. Drip tight shut-off.

h. Sealing disk: Suitable for potable water.

i. ASTM A536 ductile iron or ASTM A216 WCB cast steel.

j. Stainless steel trim.

k. Control system components to be manufactured by same company as main valve.

l. Pilot piping for the valves:

1) Design, furnish, and test per manufacturer's recommendations.

2) Hydraulic (water) operated control pilots with seamless copper or stainless steel tubing.

3) Heavy-duty, ball-type shut off valves.

4) Opening and closing speed control valves.

5) Pilot "Y" strainers with blowdown valves.

6) Check valves.

7) Atmospheric drain.

8) Pressure gauges.

9) Provide automatic vent valves.

m. Valve visual position indication:

1) Stainless steel position indicator rod.

2) Brass or stainless steel protective housing which allows viewing of valve position from two sides.

3) Manual air purge valve at top of assembly.

4) Pyrex tube or covers to enclose stem inside housing.

n. Furnish and install pilot piping and associated valving and fittings as recommended by the valve manufacturer.

2. Model 106-SC manufactured by Singer Valve, www.singervalve.com, Model No. 43-01, manufactured by Cla-Val Company, www.cla-val.com, Model A115 Series manufactured by Apollo Valves, www.apollovalves.com, Model 120-01 manufactured by OCV Control Valves, www.controlvalves.com; or equal, having the following essential characteristics:

a. Automatically controls flow through valve to specified flow rate.

b. Flow control settings shall be field adjustable.

c. Orifice plates:

1) Install orifice plates and valves as shown on the drawings. Actual valve and orifice plate locations may vary depending on control valve manufacturer's recommendations.

2) Provide orifice plate bore size in accordance with manufacturer's recommendations to meet required flow rates.

d. Solenoid shut-off valve:

1) 120 vac.

2) Control valve to close when supplied power.

3) Control valve will remain open on loss of power.

e. Limit switch:

1) Adjustable.

2) Integral to control valve.

3) Actuates when valve is fully closed.

4) Double pole double pole switches rated at 1 amp at 125 vdc and 120 vac.

5) UL rated with NEMA 4 enclosure.

B. Exposed Ball Valves:

1. Carbon Steel, Cast Iron or Bronze:

2. 3-inch and smaller:

a. 70LF-100 Series manufactured by Apollo Valves, www.apollovalves.com; or equal, having the following essential characteristics:

1) Cold water pressure rating of not less than 300 psig

2) Body: Full ported carbon steel or lead-free bronze.

3) Ball: Stainless steel or lead -free brass.

4) Valve stem: Stainless steel or lead-free brass.

5) Packing gland: Steel or brass.

6) Install where shown on drawings. Bronze or carbon steel ball valves shall be installed on carbon steel pipe.

7) Bubble tight at the rated pressure in either direction and suitable for operation after long periods of inactivity.

8) MSS SP–110.

9) Female threaded ends.

3. 4-inch and larger:

a. Val-Matic Rubber Seated Ball Valves, manufactured by Val-Matic, www.valmativ.com; or equal, having the following essential characteristics;

1) Flanged ends:

a) AWWA Class 150 valves: Flanges suitable for connection to AWWA Class D flanges or ANSI Class 150 flanges.

2) Valves suitable for manual service.

3) AWWA C507.

4) Pressure ratings suitable for steel pipe pressure.

4. Provide dielectric insulating connections as necessary to prevent galvanic corrosion between dissimilar metals.

C. Buried Ball Valves (Curb Stop):

1. Bronze:

2. 3-inch and smaller:

a. Model 76171 manufactured by A.Y. McDonald, www.aymcdonald.com; or equal, having the following essential characteristics:

1) Cold water pressure rating of not less than 300 psig

2) Body: Full ported carbon steel or lead-free bronze.

3) Ball: Stainless steel or lead -free brass.

4) Valve stem: Stainless steel or lead-free brass.

5) Packing gland: Steel or brass.

6) Install where shown on drawings. Bronze or carbon steel ball valves shall be installed on carbon steel pipe.

7) Bubble tight at the rated pressure in either direction and suitable for operation after long periods of inactivity.

8) Female threaded ends.

D. Gate Valves:

1. Series A-2361 manufactured by Mueller Co. LLC, www.muellercompany.com, High Pressure Resilient Wedge Gate Valve manufactured by Kennedy Valve, www.kennedyvalve.com; or equal, having the following essential characteristics:

a. AWWA C509 or AWWA C515.

b. OS&Y or NRS

c. Ductile iron body and wedge shall be made of ductile iron.

d. Bonnet: Ductile iron or cast iron.

e. Handwheel.

f. Valve ends shall be flat faced flanges and shall match the drilling dimensions and pattern of AWWA C207 Class F flanges.

g. Wedges shall be fully encapsulated with EPDM.

h. Minimum cold-water working pressure rating of 300 psig.

i. Zero leakage.

E. Check Valves:

1. Series 581 Silent Check Valve manufactured by Cla-Val, www.cla-val.com, or equal, having following essential characteristics:

a. Silent operating type.

b. Globe style.

c. Center guided, spring loaded disc.

d. All parts to be field replaceable.

e. Resilient seal.

f. Zero leakage.

g. Minimum working pressure rating of 300 psig, cold water service.

h. Suitable for bolting to AWWA Class F steel pipe flanges.

i. Valve body: ASTM A536 grade 65-45-12 ductile iron or ASTM A216 grade WCB cast steel.

j. Disc: ASTM A351 grade stainless-steel.

k. Seat and compression spring, type 316 stainless-steel.

l. Valve body with integral flanges (not wafer style).

m. Flat face flanges: Provide spiral or concentric serrated grooved flange face gasket surface finishes that are suitable for flat gasket material.

F. Combination Air Valves:

1. Manufactured by DeZurik/Apco, www.dezurik.com; Cla-Val, www.cla-val.com; Val-Matic Valve and Manufacturing Corporation, www.valmatic.com; or equal having the following essential characteristics:

a. Cold water service.

b. Cold-water pressure rating: 300 psig, minimum.

c. Single or dual body.

d. Body material: Ductile iron or cast steel.

e. AWWA C512.

f. Actuated by a float or floats.

g. Stainless steel internal parts such as guides, bushings, and screws.

h. Stainless steel floats.

i. Float-pivot supports: Brass, bronze, stainless steel, or cast iron.

j. Inlet: Threaded

k. Provide combination air-inlet and air-release type or air and vacuum type actuated by a float as shown on the drawings.

l. Remains open for filling line until water has displaced air at point of attachment of valve assembly after which it closes.

m. Opens when pressure in pipeline drops sufficiently to create a vacuum.

n. Releases trapped air when the pipeline is under pressure.

o. 2-inch Air release minimum orifice size: 1/16-inch.

p. 3-inch Air release minimum orifice size: 5/64-inch

q. Operating pressure for seat materials: Resilient seat suitable for 0 to 300 psig.

r. Furnish with ball valves in accordance with paragraph 2.01 F:

1) One ball valve on top, permits checking effectiveness of air valve.

2) One ball valve on bottom, allows valve to be drained.

3) For dual body valves, provide full port isolation ball valve between main body and air release valve.

G. Inflow Preventer Assembly: Val-Matic model 1302/1303 with optional bracket SPK-1302/1303-80, and field test kit SPK-1302/1303 as applicable per air valve size, manufactured by Val-Matic Valve and Manufacturing Corporation, 905 Riverside Drive, Elmhurst, IL 60126; or equal, having the following essential characteristics:

1. Manufactured in accordance with AWWA C514 and ASSE 1063.

2. Stainless steel internal parts such as guides, bushings, and screws.

3. Upper and lower chambers: Ductile iron, ASTM A536 Grade 65-45-12.

4. Basket strainer:

a. Type 304 stainless steel.

b. 24# mesh opening (0.0277 inch).

c. Quick disconnect joint.

d. Total flow area of at least four times circular cross-sectional area having a diameter equal to nominal size of air valve outlet.

5. Float checks and trim: Type 316 stainless steel.

6. Cold-water pressure rating for body: Minimum of 25 psi.

7. Resilient seat: EPDM with fiberglass reinforcement.

8. Inlet connection: Internally threaded NPT.

9. Piping manifold and wall bracket as shown on drawings.

10. Supply mounting bracket and field test kit with all appurtenant piping and valves per air valve as shown on drawings.

H. Butterfly Valves:

1. Cold water pressure up to 150 psig: AWWA C504, Class 150B.
2. Cold water pressures from 151 psig up to 250 psig: AWWA C504, Class 250B.
3. Bubble tight at rated pressure in either direction and suitable for operation after long periods of inactivity.
4. Tested in accordance with AWWA C504 including performance, leakage, and hydrostatic testing.
5. Axis of valve leaf: Horizontal when valve is in installed position except where shown on drawings.
6. Flanged ends:
   a. Class 150B valves: Body flanges suitable for connection to AWWA Class D flanges.
   b. Class 250B valves: Body flanges suitable for connection to AWWA Class F flanges.
7. Valve Seat:
   a. Rubber seat: Clamped, or mechanically secured to body or disc.
   b. Material: AWWA C504, except bonded rubber seats will not be accepted.
   c. Field replaceable seat.
   d. Located either in the valve body, mating with a stainless steel seating surface, located on valve disc or in valve disc, mating with a stainless steel seating surface located in valve body.
8. Manual Operators:
   e. Manually operated butterfly valves with worm-gear or traveling nut type manual operator in accordance with AWWA C504.
   f. Ensure rim pull and component strength meets AWWA C504.
   g. Operators that open and close valves at full unbalances pressure rating of valve.
   h. Gears of self-locking type; holds valve in all positions without fluttering or creeping.
   i. Handwheel where shown on drawings.

I. Buried butterfly and gate valves:

1. Valve boxes.

2. Square nut operator: 2-inch.

3. Provide a total of 8 steel tee-handle wrenches, with a length of 4-feet.

a. Torque rating Match or exceed maximum torque input of valve actuator.

b. Tee wrench:

1) Capable of withstanding 2 times the torque rating.

2) Painted with primer and topcoat to manufacturer's standard.

4. Steel extension stems:

a. Required to provide operating nut to within 6-inches of ground surface for buried valves:

b. Rating to match or exceed maximum torque input of valve actuator.

c. Steel centering ring with 1/8-inch minimum thickness and welded all around to top of extension stem just below the operating nut.

d. Bottom socket with mud vents, welded or pinned to provide a secure connection to the square steel extension stem.

e. Connect top nut to square steel stem with a pin or two cap screws. Capable of withstanding 2 times the torque rating.

f. Painted with primer and topcoat to manufacturer's standard.

J. Valve Boxes:

1. Model 6855 series, manufactured by Tyler Union, www.tylerunion.com; or equal, having the following characteristics:
   a. Cast iron locking lid with pentagon shaped bolt head.
   b. Cast iron extension from valve nut operator to ground surface.
   c. Cast iron collar with lid marked "WATER".
   d. 5 1/4-inch minimum inside diameter cast iron.
   e. 2-piece or 3-piece and slip type.
2. Base and adequate extension items to extend from valve nut operator to ground surface.
3. Refer to Detail A on drawing 1695-D-62549.

K. Air Vent Damper:

1. Model No. 1512, FrostSafe two-way damper manufactured by Val-Matic Valve and Manufacturing Corporation, 905 S. Riverside Drive, Elmhurst IL 60126; or equal having the following essential characteristics:
   a. For 6-inch air vent as shown on drawing 1695-D-62551.
   b. Suitable for connection between AWWA Class D flanges.
   c. Allows 2-way flow.
   d. Minimizes thermal exchange of cold and warm air in and out of valve vault.
   e. Allows vault to vent at full air valve capacity.

L. Pressure Reducing Valve:

1. Single stage and suitable for water.
2. End connections: Threaded for regulators NPS 2-inch and smaller
3. Minimum working pressure: 300 psi.

M. Hose fittings:

1. Hose Bibb:
   a. Operation: 1/4 turn.
   b. End connections: Threaded.
   c. Suitable for working pressure of pipe.
   d. Hose adapter;End connections: Threaded to hose barb.
   e. Suitable for working pressure of pipe.
2.

N. Magnetic Flowmeter:

1. Electromagnetic Flowmeter, Ultra Mag, as manufactured by McCrometer,; or equal, having the following essential characteristics;
   a. Accuracy: Plus, or minus, 0.5 percent of all flowrates, over range of measurement.
   b. +0.5% Accuracy over specified flow range:
   c. Straight pipe installation clearances:
      1) 3 diameters upstream.
      2) 2 diameters downstream.
   d. Suitable for operation in ambient temperatures ranging from -4 to 140 degrees F.
   e. Flanged, cylindrical flow-tube with polyurethane liner, grounding rings and gaskets.
   f. Flanges:
      1) Suitable for bolding to AWWA Class F.
   g. Sensor Tubes:
      1) Stainless steel electrode material
      2) Bonded epoxy liner.
   h. Transmitter:
      1) Modbus or 4-20mA HART.

2.03 APPURTENANCES:

A. Gaskets:

1. Multi-Swell Style 3760 manufactured by Garlock, www.garlock.com; or equal with the following essential characteristics:
    a. Minimum 300 psig cold water pressure rating.
    b. AWWA C207 or ASME B16.5.
    c. Flat, full face type.
    d. Suitable for cold water service.
2. Do not use compressed gaskets.

B. Delivery Point Signage Tags:

1. As shown on Drawing 1695-D-62568.
2. Engraved metal tag (blue with white lettering).
3. Wire Rope (1/8" diameter).
    a. Provide 6 spare tags, wire rope and crimps.

2.04 CONTRACTOR SOURCE QUALITY TESTING

A. Shop Testing:

1. Test valves at manufacturer's facility or an alternate facility approved by COR.
2. Tests will be Government witnessed. Notify COR 30-days prior to valve testing.
3. Minimize number of Government trips by testing valves from same manufacturer during same Government trip.
4. Prior to shop leakage test and in presence of Government, Contractor shall verify that when valve is fully closed, valve position indicator indicates, "CLOSED".
5. If indicator is not in closed position (exactly at closed), remove operator from valve, fully close valve, reinstall operator so that it indicates, "CLOSED".
6. Shop leakage test acceptance:
    a. Spray or high velocity leakage will not be acceptable.
    b. If valve leaks, adjustments shall be made until valve is drip tight with no leakage. Adjustments may include adjusting valve stops, seats, actuators, or other components.
    c. If valve is not drip tight, valve is unacceptable.
    d. After any adjustments are made, valve position indicator shall be inspected again and corrected if necessary.
7. Manually operated gate valves:

a. Leakage test in both directions in accordance with AWWA C509 or AWWA C515, as applicable. Record results.

b. Hydrostatic test in accordance with AWWA C509 or AWWA C515, as applicable. Record pressures used.

c. Pressure test valves and their seats for leaks with water in 100 percent closed position for a timed period of 5-minutes with pressure equal to rated design pressure. Valves shall be drip tight with no leakage.

d. Downstream position of valve shall be visible to Government inspector.

8. Check valves (globe style) shop testing:

a. AWWA C508 seat leakage and valve body hydrostatic test.

b. In accordance with manufacturer's recommendations.

9. Pilot-operated control valves with solenoid shutoff shop testing:

a. AWWA C530 seat leakage test and valve body hydrostatic test.

b. Manufacturers' requirements.

PART 3 EXECUTION

3.01 INSTALLATION

A. Dielectrically insulate between dissimilar metals:

1. Provide insulating type unions at threaded valve, pipe or fitting connections between dissimilar metals.

B. Provide a sufficient quantity of unions for removal, replacement and maintenance of valves and equipment.

C. Install the valves and equipment to line and grade as shown on drawings.

D. Install valves with flanged connections so that the bolt holes straddle the vertical centerline when in the installed position.

E. Install valves and gaskets in accordance with the manufacturer's instructions.

F. Valve Tags:

1. Provide brass or stainless steel tags for all valves identified in the Piping and Instrumentation Diagram drawings 1695-D-T0152 through 1695-D-T0161.

2. Stamp or engrave tag with the valve identification code designated in the Piping and Instrumentation Diagram drawings 1695-D-T0152 through 1695-D-T0161.

3. Permanently attach tag to valve with wire and metal crimp-type connector.

3.02 CONTRACTOR FIELD QUALITY TESTING

A. Field Pressure Testing:

1. After each valve has been completely installed, test valve in place by opening and closing valve through full range of operation 3 times.
2. Furnish testing equipment, as approved by COR, which provides observable and accurate measurements of leakage under specified conditions.
3. Maximum Filling Velocity: 0.25 feet per second calculated based on full area of pipe.
4. Expel air from piping system during filling.
5. Test Pressure:
    a. 150 percent of steel manifold pipe normal operating pressure.
    b. Maximum test pressure not to exceed 300 psi.
    c. Test pressure for some segments may be reduced subject to COR approval.
6. Apply and maintain specified test pressure with hydraulic force pump. Valve off piping system when test pressure is reached.
7. Maintain hydrostatic test pressure continuously for 2 hours minimum.
8. Maximum Allowable Leakage: No leakage.
9. Dispose of water used in testing.

B. Test pilot operated control valve per manufacturer's recommendation.

1. Set flow control valve full closure speed to a minimum of 60 seconds, or as close as allowed by manufacturer recommendation.

C. Test Flow control valve and Flowmeter per specification Section 01 80 15 - Commissioning.

3.03 COATINGS

A. Coat valves and equipment in accordance with Section 09 96 20 – Coatings.

END OF SECTION

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